

EXAM 7 – SPRING 2011

19. (4 points)

The Chief Risk Officer for an insurance company is evaluating various risk measures for the purpose of allocating economic capital to the two business units within the company and has computed the following at the 99% confidence level:

	<u>Business Unit 1</u>	<u>Business Unit 2</u>	<u>Total</u>
Value at Risk (VaR)	\$100,000,000	\$90,000,000	\$200,000,000
Tail Value at Risk (TVaR)	105,000,000	160,000,000	290,000,000

a. (1.5 points)

Compare in detail the relative merits of Value at Risk (VaR) versus Tail Value at Risk (TVaR).

b. (1.5 points)

Calculate the Expected Policyholder Deficit for each business unit and for the company in total.

c. (1 point)

Fully discuss one possible method to allocate economic risk capital to each business unit, including reference to a potential problem with that method.

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Sample Solutions
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Solution 1

a)

VaR

Var is not sub additive as VaR at a fixed probability level for a number of risks combined can be greater than the sum of the VaRs of the individual risks. However for many applications, sub-additivity is not necessary.

It is just a single point on the distribution so does not give very much information.

It is also difficult to allocate reasonably to business units.

“TVaR is sometimes criticized for including losses excess of insolvency, which presumably do not affect shareholders. This is not really a problem, however, as these losses can hurt policyholders, whose attitudes affect growth and profits, and hence do affect shareholders. A more serious issue with TVaR is that it is linear for losses in the excess region. This is contrary to usual risk attitudes, where a loss twice as big is regarded as more than twice as bad.”

b)

$$EPD = (1 - \alpha) \times (TVaR - VaR)$$

Unit	α	TVaR	VaR	EPD
1	0.99	105,000,000	100,000,000	50,000
2	0.99	160,000,000	90,000,000	700,000
3	0.99	290,000,000	200,000,000	900,000

c)

VaR at a high level is computed for each unit, these are added up, and each unit's share of the total is taken to be its share of the economic capital. This is in part developed from the idea that firm capital is determined by a high VaR level, so that is the risk measure to use to allocate capital to each unit.

A problem with this method is that it is not marginal, and so the risk adjusted profit calculation does not maintain the financial principle of comparing marginal profits with marginal costs.